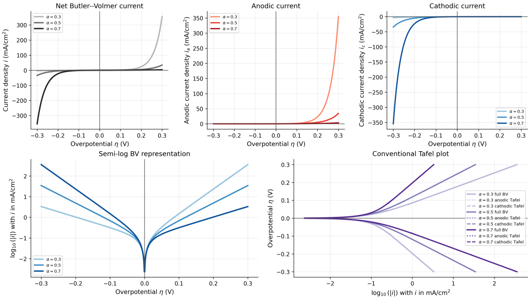


EXERCISE 4

Charge-transfer kinetics: Butler–Volmer

June 7, 2026 | O. Schmidt | Theory and Computation of Energy Materials (IET-3)

AIM OF TODAY'S EXERCISES

Today, we investigate charge-transfer kinetics using the Butler–Volmer equation. Some help can be found on

$\longrightarrow$ Exercises $\longleftarrow$

Objectives: To understand how the current density depends on

- The overpotential η
- The exchange current density i_0
- The transfer coefficient α
- The number of transferred electrons n
- The temperature T

Furthermore, we will compare the full Butler–Volmer equation with its anodic and cathodic Tafel limits.

BUTLER–VOLMER: FUNDAMENTAL CONSTANTS

Symbol	Value	Unit	Meaning
N_A	6.022×10^{23}	mol^{-1}	Avogadro constant
k_B	8.617×10^{-5}	$\mathrm{eV\,K}^{-1}$	Boltzmann constant
e_0	1.602×10^{-19}	C	Elementary charge
F	96485	$\mathrm{C\,mol}^{-1}$	Faraday constant
R	8.314	$\mathrm{J\,mol}^{-1}\,\mathrm{K}^{-1}$	Gas constant

BUTLER–VOLMER: MODEL PARAMETERS

Symbol	Unit	Meaning
η	V	Overpotential, i.e. deviation from equilibrium potential
i_0	$\mathrm{A\,m^{-2}}$	Exchange current density
α	–	Cathodic charge-transfer coefficient
$1-\alpha$	–	Anodic charge-transfer coefficient
n	–	Number of transferred electrons
T	K	Temperature
i	$\mathrm{A\,m^{-2}}$	Net current density

MODEL SETUP: CHARGE-TRANSFER KINETICS

We now move from equilibrium thermodynamics and electric-double-layer structure to charge-transfer kinetics.

In an electrochemical reaction, electrons are transferred across the electrode–electrolyte interface. The rate of this electron transfer determines the faradaic current.

The Butler–Volmer equation describes how the current density changes when the electrode potential is moved away from equilibrium.

Overpotential

$$\eta = E - E_{\mathrm{eq}}$$

At equilibrium, $\eta = 0$ and the net current density is zero.

MODEL SETUP: FORWARD AND BACKWARD REACTIONS

Consider a one-electron or multi-electron charge-transfer reaction written schematically as

$$\mathrm{Ox} + n\,\mathrm{e}^- \rightleftharpoons \mathrm{Red}.$$

Away from equilibrium, the forward and backward reaction rates are no longer equal.

Current-density decomposition

$$i(\eta) = i_\mathrm{a}(\eta) + i_\mathrm{c}(\eta)$$

$$i_\mathrm{a}(\eta) > 0, \qquad i_\mathrm{c}(\eta) < 0$$

The anodic current corresponds to oxidation, while the cathodic current corresponds to reduction.

MODEL SETUP: BUTLER–VOLMER EQUATION

The Butler–Volmer equation combines the anodic and cathodic exponential current contributions.

Butler–Volmer equation

$$i(\eta) = i_0 \left[\exp\left(\frac{(1-\alpha)nF\eta}{RT}\right) - \exp\left(-\frac{\alpha nF\eta}{RT}\right)\right]$$

Here, i_0 sets the current scale, while α controls how the overpotential is distributed between the anodic and cathodic activation barriers.

MODEL SETUP: ANODIC AND CATHODIC PARTS

The Butler–Volmer equation can be written as the sum of an anodic and a cathodic branch.

Anodic and cathodic current densities

$$i_{\mathrm{a}}(\eta) = i_0 \exp\left(\frac{(1-\alpha)nF\eta}{RT}\right)$$
$$i_{\mathrm{c}}(\eta) = -i_0 \exp\left(-\frac{\alpha nF\eta}{RT}\right)$$
$$i(\eta) = i_{\mathrm{a}}(\eta) + i_{\mathrm{c}}(\eta)$$

At $\eta = 0$, the two contributions have equal magnitude and opposite sign.

EQUILIBRIUM CONDITION

At equilibrium, the anodic and cathodic currents cancel exactly.

At equilibrium: $\eta = 0$

$$
\begin{aligned}
i_\mathrm{a}(0) &= i_0 \\
i_\mathrm{c}(0) &= -i_0 \\
i(0) &= i_\mathrm{a}(0) + i_\mathrm{c}(0) = 0
\end{aligned}
$$

The exchange current density i_0 is therefore not a net current. It is the magnitude of the anodic and cathodic currents at equilibrium.

PHYSICAL MEANING OF i_0

The exchange current density measures how fast charge transfer occurs at equilibrium.

Interpretation of i_0

- Large i_0: fast electrode kinetics.
- Small i_0: slow electrode kinetics.
- Increasing i_0 increases the magnitude of both anodic and cathodic currents.
- i_0 controls the vertical scale of the Butler–Volmer curve.

A reaction with large i_0 requires only a small overpotential to generate a large current.

PHYSICAL MEANING OF α

The transfer coefficient α controls the asymmetry between the cathodic and anodic branches.

Transfer coefficient

$$0 < \alpha < 1$$

$$\text{cathodic factor: } \alpha, \qquad \text{anodic factor: } 1 - \alpha$$

If $\alpha = 0.5$, the Butler–Volmer curve is symmetric in the anodic and cathodic directions.

If $\alpha \neq 0.5$, one branch grows more rapidly than the other.

QUESTION 1: BUTLER–VOLMER EQUATION

1.1) What is the physical meaning of the overpotential η?

1.2) Why is the net current density zero at $\eta = 0$ even though charge transfer still occurs?

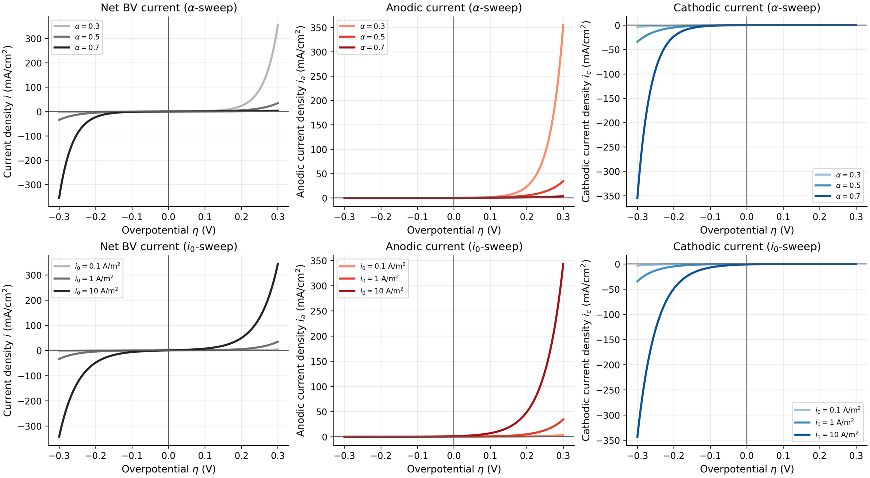
1.3) What is the meaning of the exchange current density i_0?

1.4) How does increasing i_0 change the Butler–Volmer curve?

1.5) What is the role of the transfer coefficient α?

1.6) What changes when $\alpha = 0.5$ is replaced by $\alpha \neq 0.5$?

QUESTION 1: VISUAL SOLUTION GUIDE



JÜLICH Forschungszentrum

QUESTION 1: INTERPRETATION GUIDE

What should you observe?

- At $\eta = 0$, the anodic and cathodic currents cancel.
- For $\eta > 0$, the anodic branch dominates and $i > 0$.
- For $\eta < 0$, the cathodic branch dominates and $i < 0$.
- Increasing i_0 increases the current magnitude.
- Changing α changes the asymmetry between positive and negative overpotentials.
- Increasing T makes the exponential dependence on η less steep.

MODEL SETUP: FROM BUTLER–VOLMER TO TAFEL

The full Butler–Volmer equation contains one anodic and one cathodic exponential contribution.

Butler–Volmer equation

$$i(\eta) = i_0 \left[\exp\left(\frac{(1-\alpha) nF\eta}{RT} \right) - \exp\left(-\frac{\alpha nF\eta}{RT} \right) \right]$$

For sufficiently large positive or negative overpotential, one exponential dominates. This gives the Tafel approximations.

ANODIC TAFEL LIMIT

For large positive overpotential, the anodic exponential dominates.

$$\eta \gg 0$$

Anodic Tafel approximation

$$i(\eta) \approx i_\mathrm{a}(\eta) = i_0 \exp\left(\frac{(1-\alpha)nF\eta}{RT}\right)$$

Physically, oxidation dominates and the cathodic contribution becomes negligible.

CATHODIC TAFEL LIMIT

For large negative overpotential, the cathodic exponential dominates.

$$\eta \ll 0$$

Cathodic Tafel approximation

$$i(\eta) \approx i_{\mathrm{c}}(\eta) = -i_0 \exp\left(-\frac{\alpha n F \eta}{RT}\right)$$

Physically, reduction dominates and the anodic contribution becomes negligible.

TAFEL FORM: ANODIC BRANCH

Starting from the anodic Tafel approximation,

$$i \approx i_0 \exp\left(\frac{(1-\alpha)nF\eta}{RT}\right),$$

take the logarithm:

Anodic Tafel equation

$$\eta = \frac{RT}{(1-\alpha)nF}\ln\left(\frac{i}{i_0}\right)$$

In a plot of η against $\ln(i)$, the anodic branch becomes linear.

TAFEL FORM: CATHODIC BRANCH

Starting from the cathodic Tafel approximation,

$$i \approx -i_0 \exp\left(-\frac{\alpha n F \eta}{RT}\right),$$

we use the magnitude $|i|$:

Cathodic Tafel equation

$$\eta = -\frac{RT}{\alpha n F} \ln\left(\frac{|i|}{i_0}\right)$$

In a plot of η against $\ln(|i|)$, the cathodic branch also becomes linear.

TAFEL SLOPES

The Tafel slope describes how much overpotential is needed to increase the current by one logarithmic unit.

Natural-logarithm Tafel slopes

$$b_\mathrm{a} = \frac{RT}{(1-\alpha)nF}$$
$$b_\mathrm{c} = -\frac{RT}{\alpha nF}$$

Larger n, lower T, or larger transfer coefficient in the relevant direction makes the current increase more steeply with overpotential.

TAFEL SLOPES WITH BASE-10 LOGARITHMS

Experimental Tafel plots are often shown using $\log_{10}(|i|)$ instead of $\ln(|i|)$.

Base-10 Tafel slopes

$$b_{\mathrm{a},10} = \frac{2.303RT}{(1-\alpha)nF}$$
$$b_{\mathrm{c},10} = -\frac{2.303RT}{\alpha nF}$$

These slopes have units of V per decade.

SYMMETRIC BUTLER–VOLMER CASE

A common special case is

$$\alpha = 0.5.$$

Then the anodic and cathodic transfer coefficients are equal:

$$1 - \alpha = \alpha = 0.5.$$

Symmetric Tafel slopes

$$b_{\mathrm{a},10} = -b_{\mathrm{c},10} = \frac{2.303RT}{0.5\,nF}$$

For $\alpha = 0.5$, the anodic and cathodic Tafel branches have the same absolute slope.

QUESTION 2: TAFEL APPROXIMATION

2.1) Under which condition can the anodic Tafel approximation be used?

2.2) Under which condition can the cathodic Tafel approximation be used?

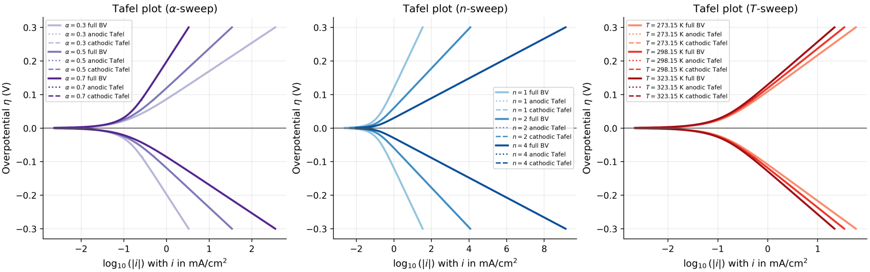
2.3) Why does the Butler–Volmer curve become approximately linear in a Tafel plot?

2.4) How does changing α affect the anodic and cathodic Tafel slopes?

2.5) How does changing T affect the Tafel slope?

2.6) How does changing n affect the Tafel slope?

QUESTION 2: VISUAL SOLUTION GUIDE



JÜLICH Forschungszentrum

MODEL ASSESSMENT: BUTLER–VOLMER

The Butler–Volmer equation provides a simple kinetic relation between overpotential and faradaic current density.

It captures the competition between oxidation and reduction through the anodic and cathodic exponential terms.

Strength of the model

$$i(\eta) = i_0 \left[\exp\left(\frac{(1-\alpha)nF\eta}{RT}\right) - \exp\left(-\frac{\alpha nF\eta}{RT}\right)\right]$$

The model is especially useful because it connects measurable current densities to kinetic parameters such as i_0 and α.

STRENGTHS OF BUTLER–VOLMER

What does Butler–Volmer describe well?

- The net current is zero at equilibrium.
- The current increases exponentially with overpotential.
- The anodic and cathodic directions can be treated in one equation.
- The exchange current density i_0 gives a measure of kinetic activity.
- The transfer coefficient α describes asymmetry between oxidation and reduction.
- At large overpotentials, the model reduces to the Tafel approximation.

LIMITATIONS OF BUTLER–VOLMER

The basic Butler–Volmer equation describes charge-transfer kinetics, but it does not contain all interfacial effects.

What is missing?

- It does not explicitly describe mass transport limitations.
- It does not include concentration gradients near the electrode.
- It does not describe electric-double-layer structure.
- It assumes that i_0 and α are known parameters.
- It does not include specific adsorption or surface restructuring.
- It is a kinetic boundary relation, not a full transport model.

CONNECTION TO PREVIOUS EXERCISES

In the previous exercises, we studied equilibrium thermodynamics and electric-double-layer models.

Butler–Volmer adds the kinetic part: it describes how fast charge is transferred once an overpotential is applied.

Conceptual connection

- Nernst equation: equilibrium potential.
- Gouy–Chapman / Stern / Bikerman: interfacial electrostatic structure.
- Butler–Volmer: faradaic current generated by charge transfer.

A complete electrochemical model often needs all three ingredients: thermodynamics, double-layer structure, and kinetics.

QUESTION 3: OVERALL MODEL ASSESSMENT

3.1) What are the strengths of the Butler–Volmer equation?

3.2) What are the main limitations of the Butler–Volmer equation?

3.3) Why is Butler–Volmer not a complete mass-transport model?

3.4) Which parameters control the kinetic response of the electrode?

3.5) How does Butler–Volmer connect to the Nernst equation?

3.6) How does Butler–Volmer connect to electric-double-layer models?

QUESTION 3: INTERPRETATION GUIDE

What should you conclude?

- Butler–Volmer is a compact model for charge-transfer kinetics.
- It explains why the current vanishes at equilibrium.
- It predicts exponential current growth away from equilibrium.
- It connects naturally to Tafel analysis at large overpotentials.
- It must be coupled to transport and double-layer models when concentration gradients or interfacial structure are important.

JÜLICH Forschungszentrum